Refined Reconstruction

$X_{refined} \in R^{B \times L \times C}$

X_1 X_2 X_3 ⋮ X_C

= observed patch (fully observed)

= recovered patch (previously missing)

= mixed patch (observed + filled)

Observed-value Preservation

Network Residual Head

Gated Reference Correction

Observation-Guided Residual Refinement

Initial Reconstruction

Confidence-Weighted Overlap Fusion

Pointwise Evidence-Guided Reconstruction

Patch Evidence Bank

z_0

Patch Embedding

LayerNorm

Point-level Context Encoding

Observation Normalization

Observed Sequence and Mask

(X_{obs}, M)

X_1 X_2 X_3 ⋮ X_C

= observed patch (fully observed)

= missing patch (fully missing)

= partial patch (partly observed)

(d) Observation-Guided Residual Refinement

Refinement Features

FFN

LayerNorm

Temporal attention weights

MatMul

Scale & Softmax

Value

Key

Query

LayerNorm

MatMul

t_{i-1} t_i t_{i+1} t_{i+2}

Encoder

(b) Multi-source Patch Evidence Construction

X_a X_c X_b

a1 a2 c1 Target c2 b1

Time

T: Temporal Retrieval — c1 c2 — Patch Encoder — C_T

V: Variable Retrieval — a1 a2 b1 — Patch Encoder — C_V

H: Memory Retrieval — h1 h2 — Patch Encoder — C_H

Target — Historical Patch Retrieval — h1 h2 h3

(c) Pointwise Evidence-Guided Reconstruction

$F^{(2)}$ z_0 Position

Query — Target Context Encoding — Base Reconstruction Head

MatMul — Scale & Softmax — Point-Patch weights

Key Value — MatMul

C r — Concat — Joint Residual Head

Δx

(a) Point-level Context Encoding

Time

Variate

Query Key Value — MatMul — Scale & Softmax — Variable attention weights

MatMul

LayerNorm

FFN

$F^{(2)}$

X_c

t_{i-1} t_i t_{i+1} t_{i+2}

Encoder

$F^{(1)}$